

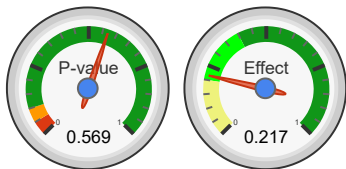
Statistics Kingdom

Reporting results in APA Format

Results of the D'Agostino-Pearson test indicated that there is a non significant difference from the normal distribution , ($\chi^2(2) = 1.13, p = .569$).

How to do with R?

Parameter	Value
P-value	0.5694
χ^2	1.1265
Sample size (n)	24
Average ($\bar{x}$)	59.75
Median	58.5
Sample Standard Deviation (S)	19.8762
Sum of Squares	9086.5
Skewness	0.1957
Skewness Shape	 Potentially Symmetrical (pval=0.679)
Excess kurtosis	-0.7855
Kurtosis Shape	 Potentially Mesokurtic , normal like tails (pval=0.392)
Outliers	



D'Agostino-Pearson test test, using Chi-Square distribution (right-tailed)[\[Validation\]](#)

1. H₀ hypothesis

Since $p\text{-value} > \alpha$, we accept the H_0 .
It is assumed that the data distribution is normal.

In other words, the difference between the data sample and the normal distribution is not big enough to be statistically significant.
A non-significance result can not prove that H_0 is correct, only that the null assumption can not be rejected.

2. P-value

The p-value equals **0.5694**, ($P(x \leq 1.1265) = 0.4306$). It means that the chance of type I error, rejecting a correct H_0 , is too high: 0.5694 (56.94%). The larger the p-value the more it supports H_0 .

3. Test statistic

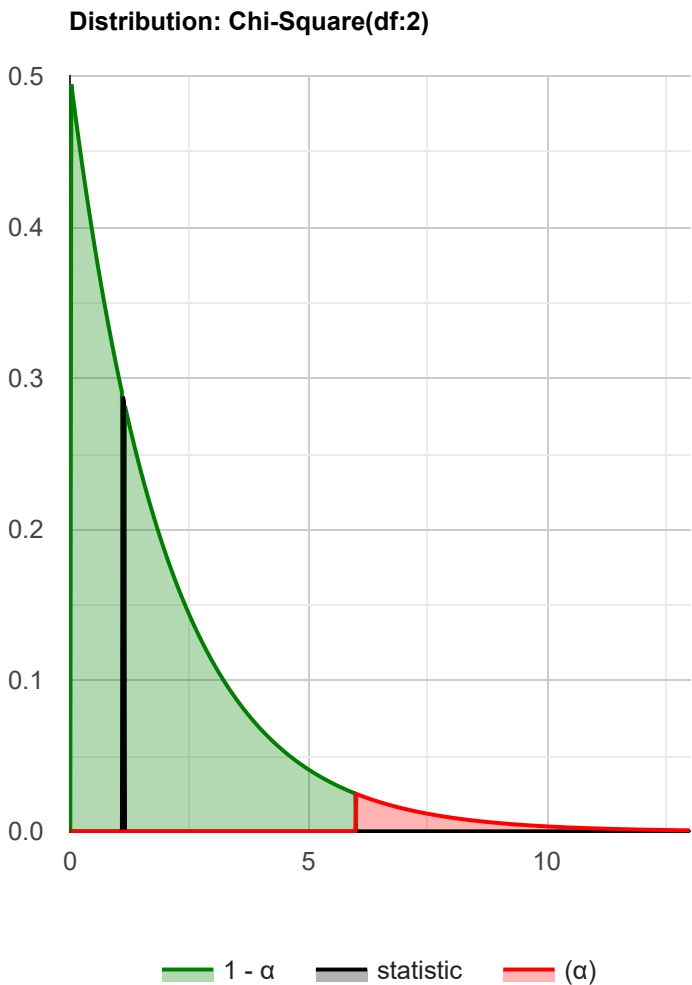
The test statistic χ^2 equals **1.1265**, which is in the 95% region of acceptance: $[-\infty : 5.9915]$.

4. Effect size

The observed effect size φ is **small**, **0.217**. This indicates that the magnitude of the difference between the sample distribution and the normal distribution is small.

Since the null assumption cannot be rejected, you may ignore the effect size.

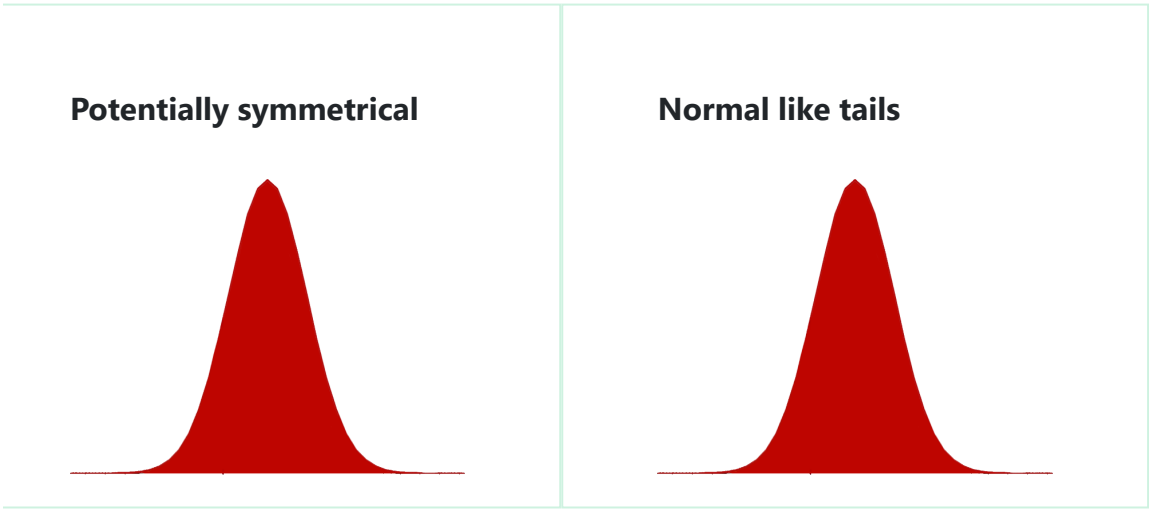
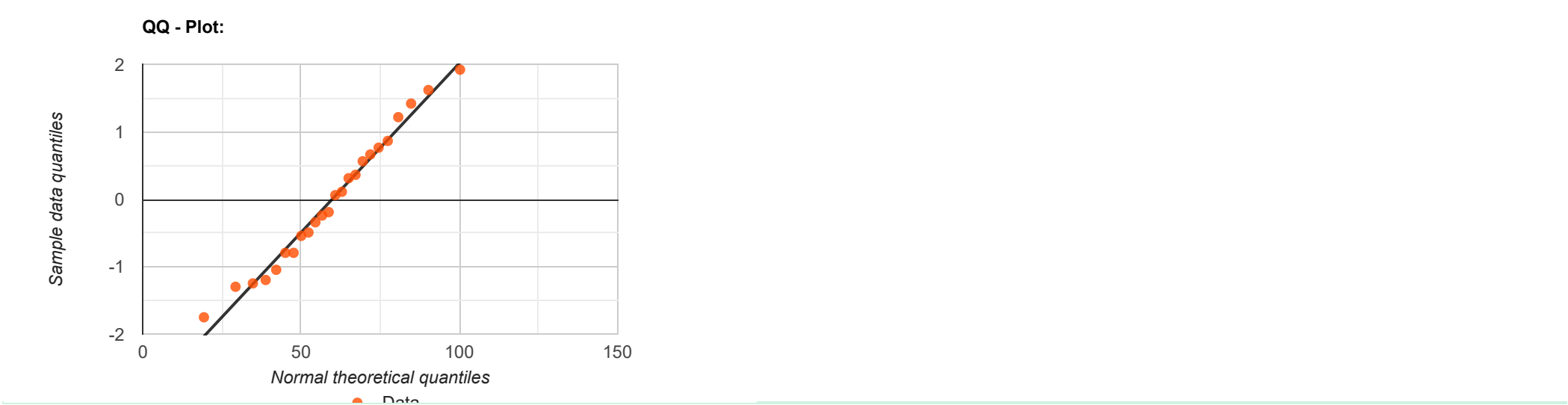
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Validation

- Outliers**
[Outliers'](#) detection method: Tukey Fence, $k=1.5$.
The data doesn't contain outliers.

- Sample size**
The sample size is 24.



$$\text{skewness} = \frac{\sum (x_i - \bar{x})^3 / n}{(\sum (x_i - \bar{x})^2 / n)^{1.5}} = 0.1833$$

$$y = \text{skewness} * \sqrt{\left(\frac{(n + 1)(n + 3)}{6(n - 2)}\right)} = 0.4144$$

$$b = \frac{3(n^2 + 27n - 70)(n + 1)(n + 3)}{(n - 2)(n + 5)(n + 7)(n + 9)} = 3.5804$$

$$w^2 = \sqrt{(2(b - 1))} - 1 = 1.2718$$

$$d = \frac{1}{\sqrt{(\ln(w))}} = 2.8844$$

$$a = \sqrt{\left(\frac{2}{w^2 - 1}\right)} = 2.7129$$

$$Z_{\text{skewness}} = d * \ln[y/a + \sqrt{(y/a)^2 + 1}] = 0.4389$$

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$$\text{kurtosis} = \frac{\sum (x_i - \bar{x})^4 / n}{(\sum (x_i - \bar{x})^2 / n)^2} = 2.1289$$

$$E(\text{kurtosis}) = \frac{3(n - 1)}{n + 1} = 2.76$$

$$\text{Var(kurtosis)} = \frac{24n(n - 2)(n - 3)}{(n + 1)^2(n + 3)(n + 5)} = 0.5438$$

$$x = \frac{\text{kurtosis} - \text{E(kurtosis)}}{\sqrt{(\text{Var(kurtosis)})}}$$

$$B = \frac{6(n^2 - 5n + 2)}{(n + 7)(n + 9)}\sqrt{(\frac{6(n + 3)(n + 5)}{n(n - 2)(n - 3)})} = 1.7485$$

$$A = 6 + \frac{8}{B} \left[-\frac{2}{B} + \sqrt{\left(1 + \frac{4}{B}\right)} \right] = 18.1847$$

$$\text{pos} = \left(\frac{1 - 2/A}{1 + x\sqrt{2/(A - 4)}}\right)^{1/3} = 1.0946$$

$$Z_{\text{kurtosis}} = \frac{1 - 2/(9A) - \text{pos}}{\sqrt{(2/(9A))}} = -0.9663$$

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$$\chi^2(2) = Z_{\text{kurtosis}}^2 + Z_{\text{skewness}}^2$$

$$\chi^2(2) = -0.9663^2 + 0.4389^2 = 1.1265$$

$$p = p(x \leq 1.1265) = 0.4306$$

$$\text{p-value} = 1 - p = 1 - 0.4306 = 0.5694$$
